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"AN ENERGY-AWARE AGENTIC GEO-AI FRAMEWORK FOR ENVIRONMENTAL MONITORING USING LIGHTWEIGHT SIMULATION"

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OUTLINE

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Introduction

- Geospatial Artificial Intelligence (Geo-AI) plays an increasingly important role in environmental monitoring by enabling spatial coverage analysis, risk assessment, and autonomous operation in complex environments *(Janowicz et al., 2020)*. As monitoring systems become more autonomous, they must function under practical constraints such as limited energy resources, partial observability, and dynamic environmental conditions.

- Most existing Geo-AI approaches emphasize data-driven prediction rather than explicit decision-making under constraints. This work presents a lightweight, decision-centric Geo-AI framework that focuses on energy-aware reasoning through simulation. By using utility-based action selection and interpretable agent behavior, the proposed approach enables efficient and sustainable environmental monitoring without relying on computationally intensive end-to-end learning methods.

Motivation

Autonomous environmental monitoring systems must operate for extended periods under strict energy constraints while maintaining effective spatial coverage and avoiding risky regions. Many existing Geo-AI approaches rely on heavy learning models and black-box decision policies, which increase computational cost and reduce interpretability. This work is motivated by the need for a lightweight, transparent, and energy-aware decision-making framework that enables sustainable environmental monitoring through explicit reasoning rather than end-to-end policy learning.

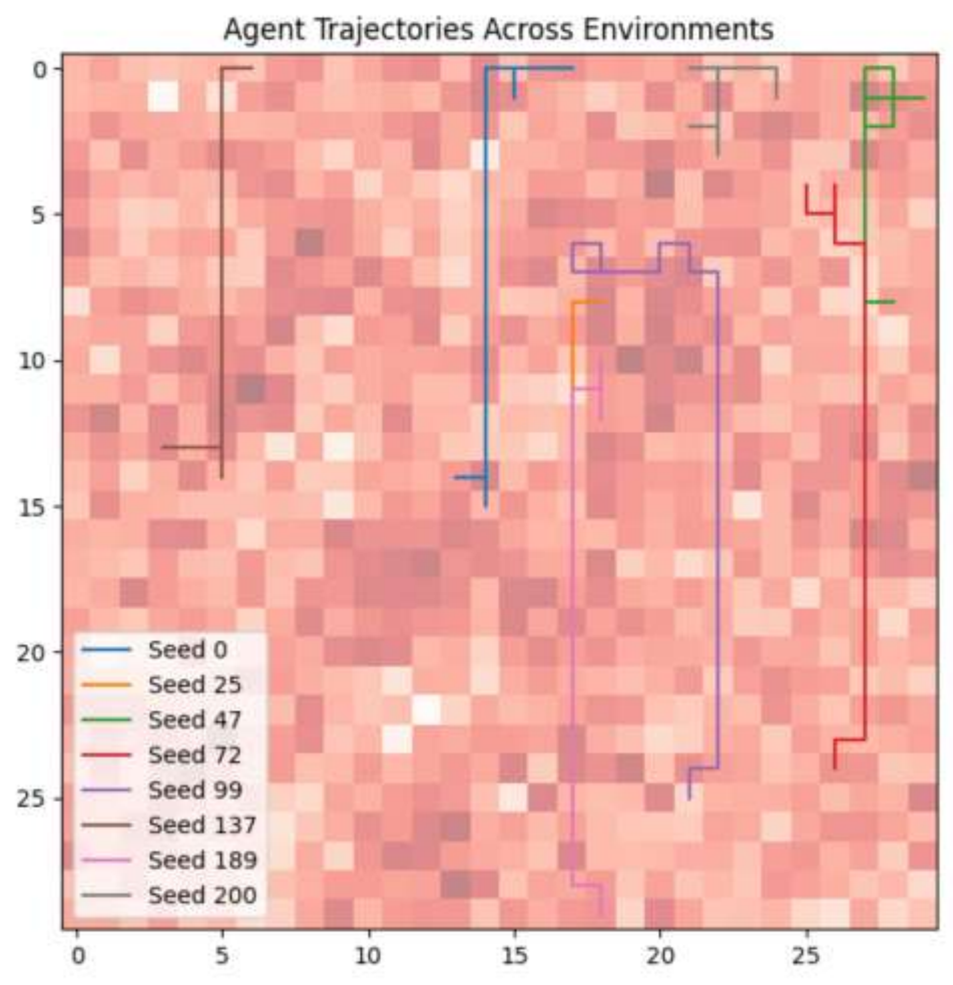
Problem Statement

- Environmental monitoring agents operate in spatial environments where energy availability, environmental risk, and coverage requirements vary over time.
- Decision-making in such environments is challenging due to partial observability and dynamically changing terrain and risk conditions.
- Many existing approaches do not explicitly model the trade-offs between energy consumption, coverage effectiveness, and environmental risk during agent operation.
- As a result, sustaining effective monitoring performance over long durations under limited resources remains challenging.

LITERATURE REVIEW

- Prior research in autonomous systems and robotics highlights the challenges of decision-making under uncertainty, limited resources, and spatial constraints in dynamic environments *(Thrun et al., 2002; Russell & Norvig, 1995)*.
- Recent Geo-AI studies focus on integrating artificial intelligence with geospatial data for environmental monitoring, emphasizing predictive modelling and spatial representation rather than explicit decision-making under operational constraints *(Janowicz et al., 2020).*
- Energy-aware navigation and path-planning methods address resource limitations but are commonly formulated as optimization problems, with limited consideration of sequential trade-offs between coverage, risk, and energy *(Zhang et al., 2021)*.
- Reinforcement learning approaches have shown strong performance in autonomous exploration; however, they often require extensive training and result in policies that lack interpretability in energy-constrained settings *(Sutton & Barto, 1998; Kober et al., 2013).*

Methodology



- **Geo-Environment Representation**
  - The monitoring region is modeled as a **two-dimensional grid-based geo-environment**, as shown in the figure, where each cell represents a spatial location with associated environmental properties.
  - The background heatmap depicts the **spatial distribution of environmental risk**, with darker regions indicating higher hazard intensity that evolves dynamically over time.
  - **Terrain-dependent energy cost** is encoded implicitly in the environment and influences agent movement, causing detours and direction changes in regions with higher traversal cost.
  - **Agent trajectories** overlaid on the risk map illustrate how movement decisions are shaped by local risk, energy considerations, and coverage dynamics rather than geometric shortest paths.
  - Repeated traversal through certain regions reflects the effect of **coverage decay**, which encourages the agent to revisit previously monitored areas as information becomes outdated.

- **<u>Agent Design and Decision Logic</u>**
  - The monitoring agent is designed as an autonomous, decision-making entity operating under partial observability within the simulated geo-environment.
  - At each time step, the agent perceives a local state consisting of its current position, remaining energy, nearby environmental risk, terrain-dependent movement cost, and coverage information.
  - Based on the observed state, the agent generates a discrete set of candidate movement and sensing actions.
  - A rule-based feasibility layer filters candidate actions to ensure operational safety, energy availability, and compliance with environmental risk limits.
  - Only feasible actions are forwarded to the decision stage, ensuring that all agent behavior remains safe, interpretable, and physically plausible.

Utility-Based Reasoning and Outcome Learning

- Agent decisions are guided by a utility-based reasoning framework that explicitly balances competing monitoring objectives.
- For each feasible action a, the agent computes a scalar utility value:

$$U(a) = w_c\, C(a) - w_e\, E(a) - w_r\, R(a)$$

- Here, $C(a)$ represents the expected coverage improvement, $E(a)$ denotes the estimated energy consumption, and $R(a)$ captures the associated environmental risk.
- The weights w_c, w_e, w_r encode operational priorities and enable transparent trade-offs between coverage, energy efficiency, and risk awareness (*Keeney & Raiffa, 1993; Deb, 2011)*.
- Outcome learning is realized through lightweight supervised machine learning models trained online using accumulated state–action–outcome tuples during agent interaction with the environment.
- Interpretable models such as linear regression, decision trees, and random forests (*Breiman, 2001; Quinlan, 2014; Hastie et al., 2009)* are used to estimate expected coverage gain, energy consumption, and environmental risk, and these predictions are incorporated into the utility computation without overriding rule-based constraints or directly selecting actions.

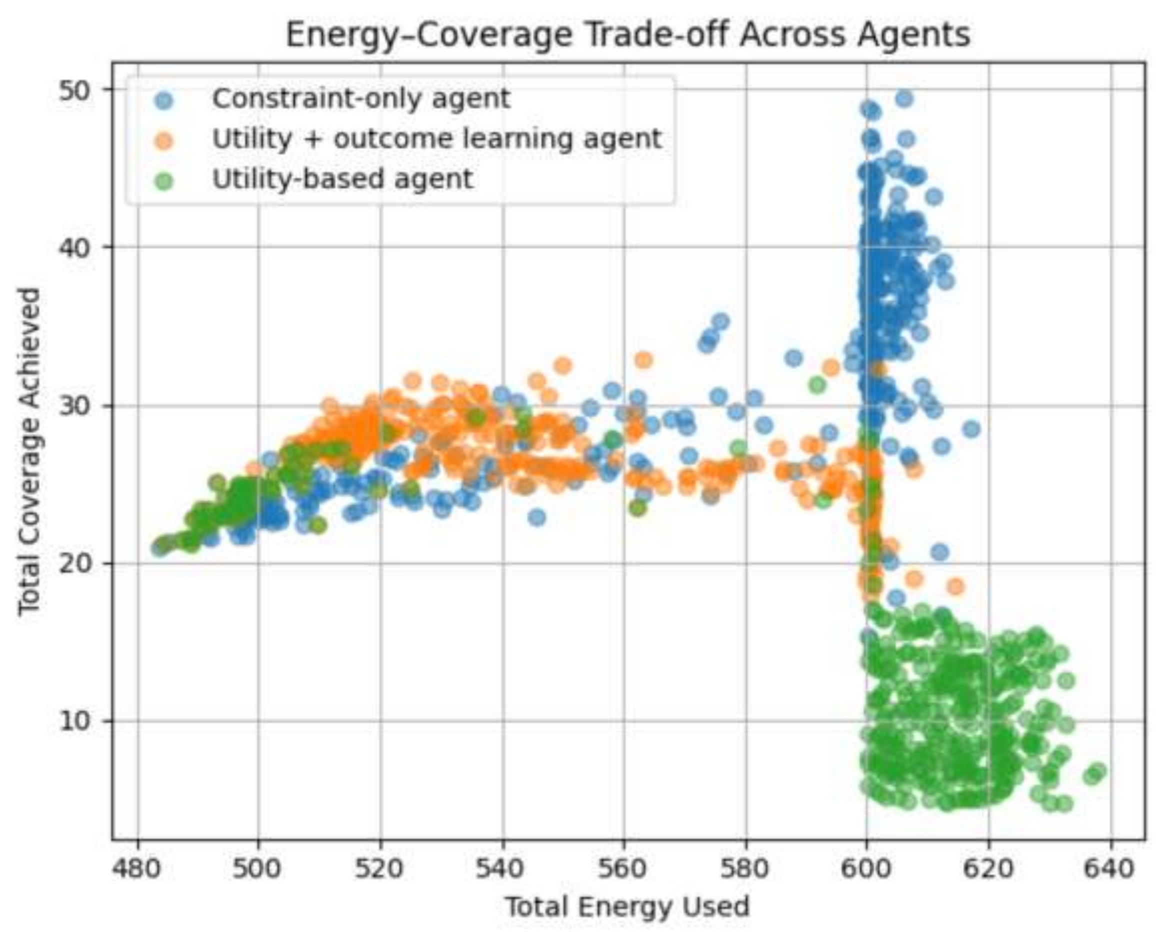
Experimental Setup and Evaluation Metrics

- Experiments are conducted in a controlled, lightweight simulation environment with randomly generated spatial risk fields, terrain energy costs, and initial agent positions.
- Multiple independent runs are performed using different random seeds to evaluate robustness across diverse environmental realizations.
- Three agent configurations are evaluated under identical conditions:
  - **Constraint-only agent**
  - **Utility-based agent**
  - **Utility-based agent with outcome learning**
- Performance is evaluated using the following metrics:
  - **Total energy consumption**, measuring resource efficiency
  - **Accumulated spatial coverage**, measuring monitoring effectiveness
  - **Sustainability index**, defined as the ratio of coverage to energy consumed
  - **Robustness**, assessed through performance stability under varying environmental volatility

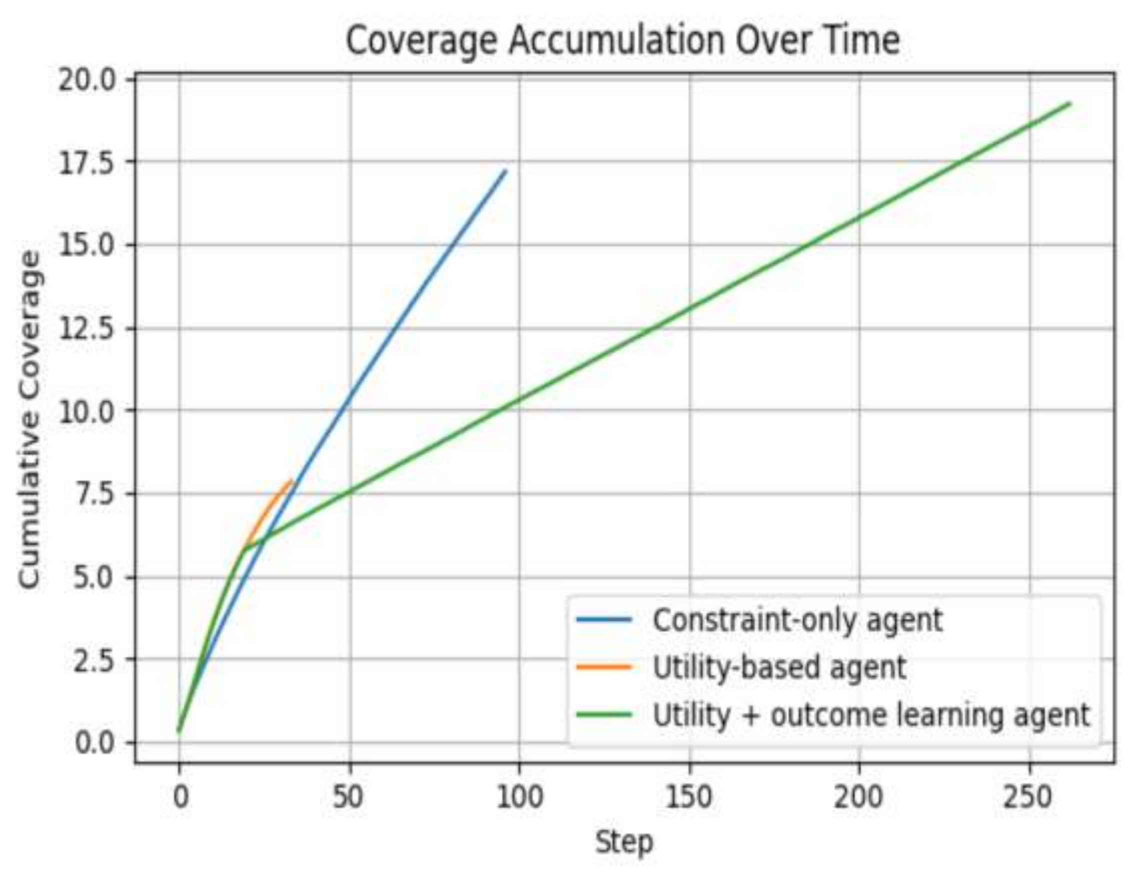
Results

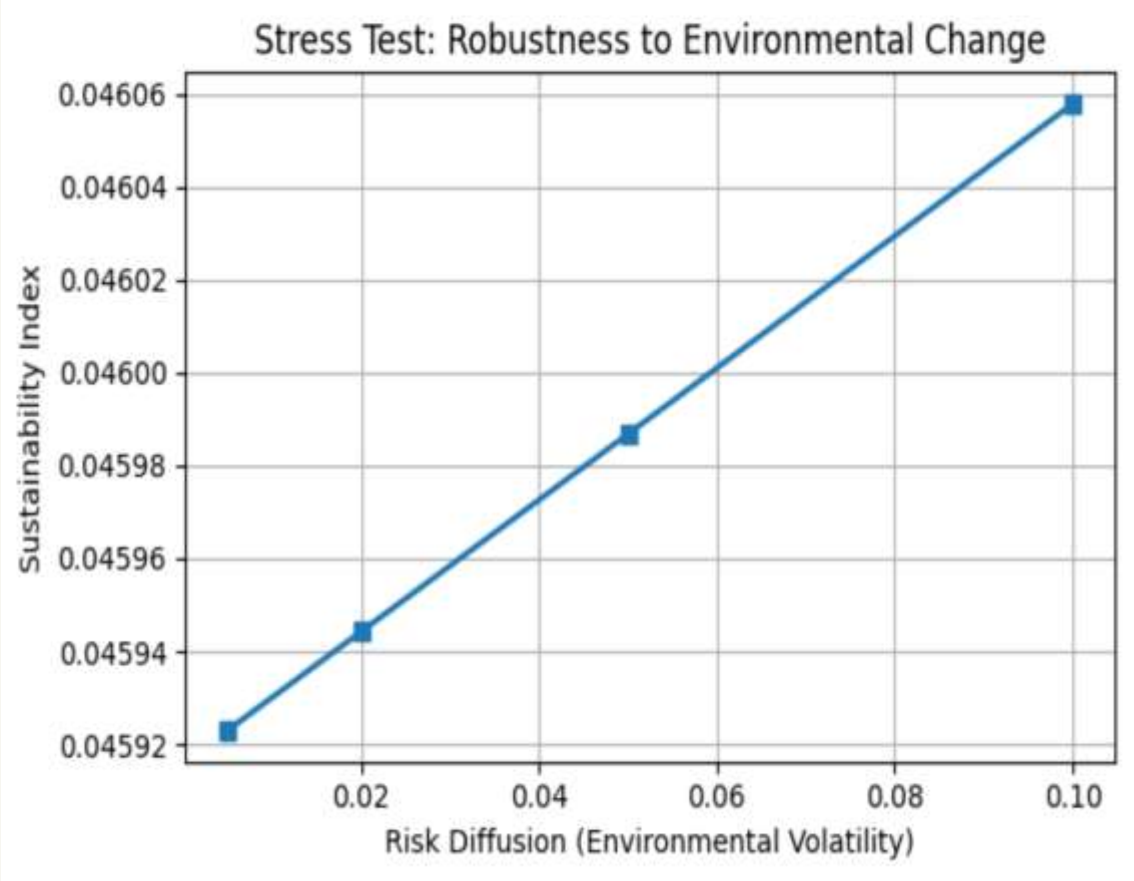
Table 1. Agent-level performance summary (mean values across trials).

Agent	Energy Used	Coverage	Sustainability
Constraint-only agent	575.90	32.46	0.056
Utility-based agent	594.26	13.18	0.023
Utility + outcome learning agent	540.18	26.39	0.049



- **Coverage Accumulation and Stress Test**





Conclusion and Future Work

- This work presented a lightweight, decision-centric Geo-AI framework for energy-aware environmental monitoring that emphasizes explicit reasoning under spatial, energetic, and risk constraints. Through a two-dimensional simulation environment with risk, terrain-dependent energy cost, and coverage dynamics, the framework enables interpretable and computationally efficient analysis of autonomous monitoring behavior without relying on end-to-end policy learning. Experimental results demonstrate that integrating utility-based reasoning with lightweight outcome learning achieves a more balanced trade-off between energy consumption and coverage, leading to improved sustainability and robustness compared to rule-based and utility-only baselines.
- Future work will extend the proposed framework to more realistic settings by incorporating three-dimensional environments with elevation, wind fields, and volumetric obstacles. Additional directions include multi-agent coordination under shared energy constraints, integration with higher-fidelity simulators or real-world datasets, and exploration of sim-to-real transfer, while preserving the decision-centric and interpretable nature of the methodology.

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